

Improving Low-Precision Network Quantization via Bin Regularization

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Abstract

Model quantization is an important mechanism for energy-efficient deployment of deep neural networks on resource-constrained devices by reducing the bit precision of weights and activations. However, it remains challenging to maintain high accuracy as bit precision decreases, especially for low-precision networks (e.g., 2-bit MobileNetV2). Existing methods have been explored to address this problem by minimizing the quantization error or mimicking the data distribution of full-precision networks. In this work, we propose a novel weight regularization algorithm for improving low-precision network quantization. Instead of constraining the overall data distribution, we separately optimize all elements in each quantization bin to be as close to the target quantized value as possible. Such bin regularization (BR) mechanism encourages the weight distribution of each quantization bin to be sharp and approximate to a Dirac delta distribution ideally. Experiments demonstrate that our method achieves consistent improvements over the state-of-the-art quantization-aware training methods for different low-precision networks. Particularly, our bin regularization improves LSQ for 2-bit MobileNetV2 and MobileNetV3-Small by 3.9% and 4.9% top-1 accuracy on ImageNet, respectively.

1. Introduction

Deep Convolutional Neural Networks (CNNs) have achieved remarkable progress in a wide range of applications including computer vision, natural language processing, speech recognition, etc. With the popularity of CNNs, there is an increasing demand for techniques to run networks efficiently on resource-constrained devices (e.g., mobile phones or FPGA). These techniques include network quantization, pruning, manual design of efficient architecture, or neural architecture search (NAS). We focus on network quantization by quantizing the networks to low bit precision in this work. It allows reduced memory footprint, faster inference time and lower power consumption when

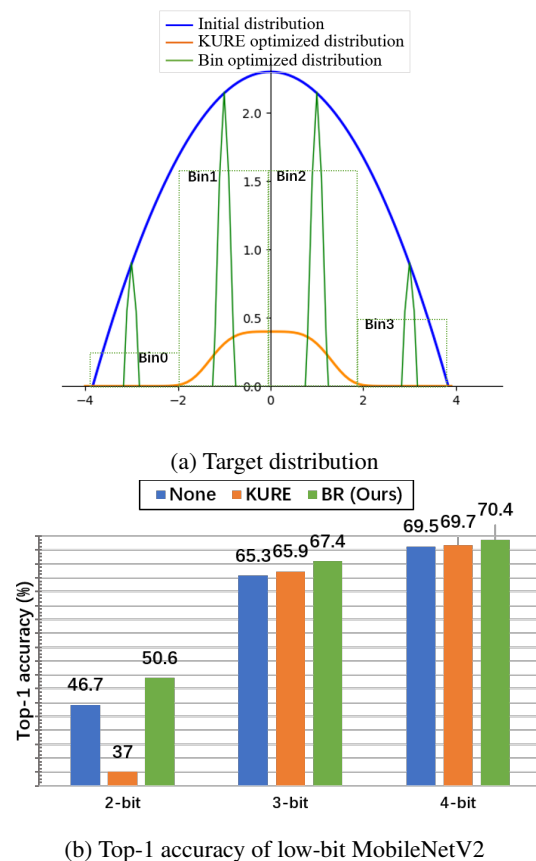


Figure 1: (a) Illustration of target distributions by different regularization methods. KURE [31] encourages the overall distribution to be uniform, while our bin regularization (BR) encourages each quantization bin distribution to be sharp. (b) Top-1 accuracy of low-bit MobileNetV2 with different regularization methods.

deploying CNN models on the edge devices.

Usually, the network accuracy decreases and the hardware performance increases as bit precision decreases. Recent network quantization methods have shown promising performance for 8-bit quantization by post-training quanti-